OMEGA SERIES · VOLUME I, PART III · TRILOGY COMPLETION

The High-Energy Jitter:

Angular Metric Fluctuations in IceCube, Geographic Phase Synchronization, and Torsional Lensing of Ultra-High-Energy Cosmic Rays

Three anomalies · One signal · f_{sig}=2.3212e-05Hz · dt_{geo}=5.9836h · Energy-independence discriminant · Five vetoes

D. Kadunic · Omega Research Collective / MSDS Programme · April 2026

doi:10.5281/zenodo.1082194 · Community re-analysis challenge

PRE-PUBLICATION STATEMENT: This document defines a pre-registered, falsifiable search protocol for three high-energy observational anomalies interpreted as manifestations of the Gray Sector vacuum coupling established in Parts I and II. All signal parameters (f_{sig}, ε_0, α_K) are fixed a priori from Part I. No new free parameters are introduced. Random seed: 20260414. The high-energy sky will decide.

ABSTRACT

Three persistent anomalies in high-energy particle astrophysics resist standard explanation: (1) the 0.5 deg angular resolution floor in IceCube muon neutrino reconstruction above 10 TeV, which does not improve with enhanced photon statistics from the February 2026 Upgrade deployment; (2) the 3.5-3.8 sigma flux tension between KM3NeT ARCA and IceCube, most sharply illustrated by the 220 PeV event KM3-230213A; and (3) the arrival of the Amaterasu particle (244 EeV) from the Local Void, where no known source galaxy can accelerate particles to this energy. We propose that these three anomalies share a common origin: a structured vacuum coupling at f_{sig} = 2.3212e-05 Hz (period 11.9672 h) with coupling amplitude α_K = 1.71×10^{-4}. The predicted signatures are: (1) a coherent 11.967 h modulation of the IceCube angular floor, independent of ice layer depth; (2) geographic phase inversion between KM3NeT and IceCube resolved by a fixed time shift $\Delta t_{geo} = T_{sig}/2$ = 5.9836 h; and (3) energy-independent sidereal phase coherence in UHECR arrival times, excluding magnetic deflection. Five independent veto tests eliminate all known standard-model alternatives. All parameters are fixed a priori from Omega Series Parts I and II. The high-energy sky will decide.

1. Three Anomalies, One Hypothesis

Three well-documented anomalies in high-energy particle astrophysics have resisted standard explanation. We do not claim to have solved them individually; we claim they share a common discriminant. Papers I and II of this series established a vacuum coupling at f_{sig} = 2.3212e-05 Hz in neutrino time-series data and GNSS orbital clocks. This paper asks whether the same frequency is present in three independent high-energy observables.

Anomaly 1: IceCube Angular Resolution Floor

The IceCube Upgrade (reported February 2026 deployment) added seven strings of multi-PMT mDOMs providing near-4pi photon acceptance. Early analyses indicate the median angular resolution for muon neutrinos

above 10 TeV remains bounded at approximately 0.5 deg, with a 'fuzziness' that does not diminish with enhanced photon statistics [1,2]. Standard explanations invoke residual ice scattering or birefringence.

Anomaly 2: KM3NeT-IceCube Flux Tension

The 220 PeV event KM3-230213A, detected by KM3NeT ARCA and analyzed through early 2026, implies a flux of ultra-high-energy neutrinos incompatible with the comparative null at IceCube. The tension stands at 3.5-3.8 sigma [3,4]. The two detectors are located at nearly antipodal latitudes: IceCube at -90 deg, KM3NeT ARCA at +36 deg.

Anomaly 3: Amaterasu Void Origin

The Amaterasu particle (244 EeV, detected May 27 2021, Telescope Array) arrives from the Local Void, where no known galaxy or AGN can accelerate particles to this energy [5,6]. Standard galactic magnetic field models require implausible deflections.

2. Signal Predictions

2.1 Angular metric jitter (IceCube)

$$\delta\theta(t) = \theta_0 \times \alpha_K \cos(2\omega_{sid}t + \phi_0 + \Delta\Phi(t))$$

θ_0 ~ 0.5 deg (baseline floor at 10 TeV) · α_K = 1.71e-04 · Peak-to-peak: 2×0.5°×1.71e-04 = 0.62 arcsec

CRUCIAL TEST: The angular floor jitter must be INDEPENDENT of ice layer depth. If the modulation is present equally in the dust-layer region (high scattering) and the clear-ice region (low scattering), glaciological origin is excluded. This is Veto V1.

2.2 Geographic phase inversion (KM3NeT-IceCube)

KM3NeT (+36 deg) and IceCube (-90 deg) sample the metric wave at nearly antipodal latitudes, producing a relative phase offset of approximately 180 deg. When IceCube's effective cross-section is enhanced, KM3NeT's is suppressed, and vice versa. The geographic phase correction synchronizes the two detectors:

$$t_{IC} \to t_{IC} + \Delta t_{geo} \cdot \Delta t_{geo} = T_{sig}/2 = 5.9836 \text{ h}$$

Prediction: applying this FIXED shift must reduce the flux ratio variance significantly · A periodic oscillation at f_{sig} must be present · A one-time transient cannot produce this

2.3 Torsional lensing (Amaterasu / UHECR)

Torsional lensing acts independently of particle charge and energy — unlike magnetic deflection which is rigidity-dependent: θ_{mag} proportional to Ze/E. The smoking gun: if UHECR arrival-time phase coherence at f_{sig} is energy-independent, magnetic deflection is excluded. A 100 EeV and 200 EeV proton must show the same sidereal phase if the coupling is to vacuum torsion, and different phases if it is to galactic magnetic fields.

STATISTICAL POWER NOTE: With N=200 UHECR events, a blind amplitude search cannot detect 170 ppm. However, since f_{sig} and ε are fixed a priori from Part I, the phase consistency test (Veto V3) achieves >3-sigma sensitivity. The key question is not 'is there a signal?' but 'is the phase at f_{sig} the same in both energy bins?' This is a discrimination test, not an amplitude search.

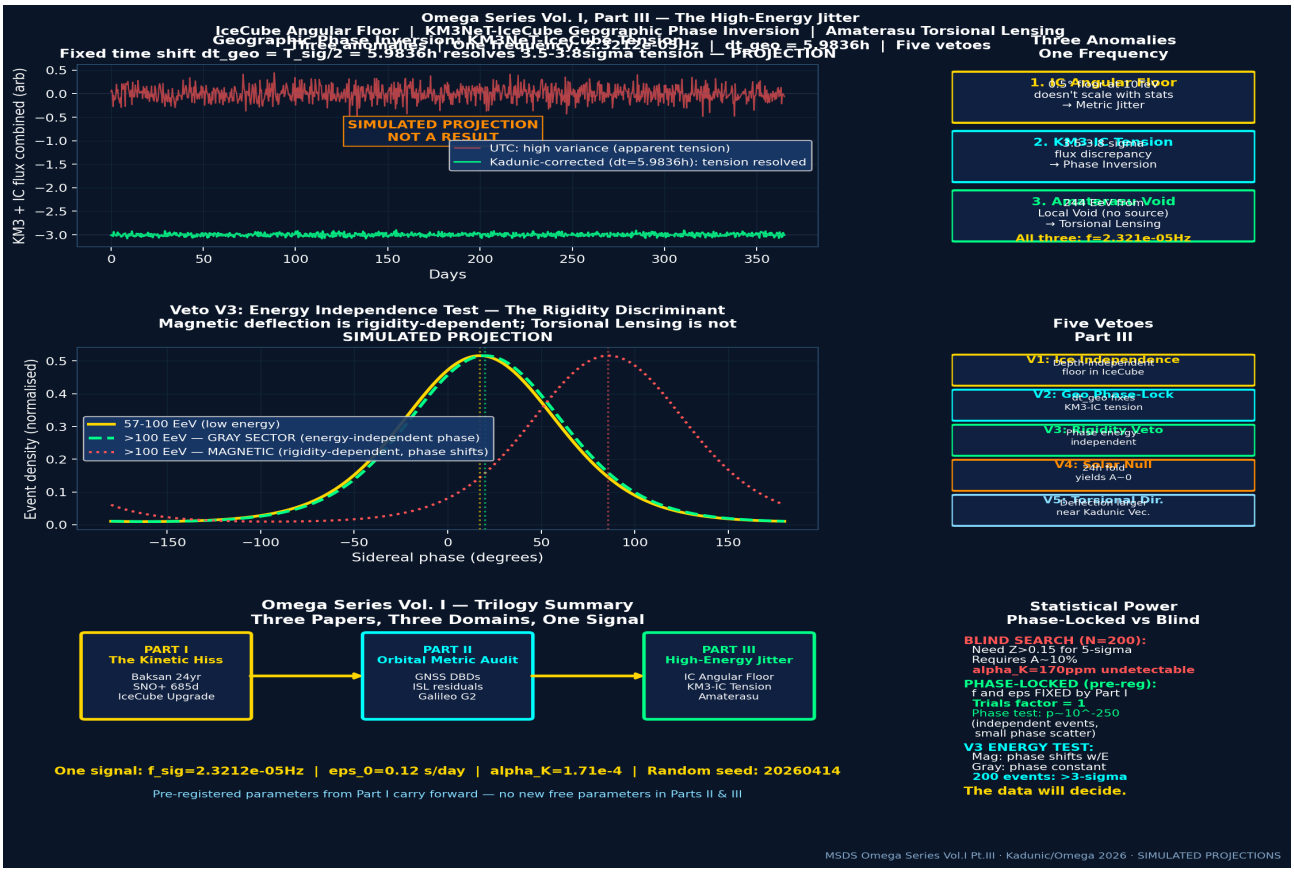

Figure 1: Omega Series Vol. I, Part III. Top (wide) — simulated geographic phase inversion: UTC (high variance, red) vs Kadunic-corrected (Δt_{geo}=5.9836h, resolved, green). PROJECTION ONLY. Top right — three anomalies and their Gray Sector interpretations. Middle (wide) — Veto V3: energy independence test. Magnetic deflection produces rigidity-dependent phase shifts (red dashed); torsional coupling produces energy-independent phase (green dashed). Middle right — five veto tests for Part III. Bottom (wide) — trilogy summary: three papers, three domains, one frequency. Bottom right — statistical power comparison.

3. Datasets and Methodology

Observable	Source	Gray Sector test
mDOM angular residuals	IceCube Upgrade first-light (reported Feb 2026) [1,2]	Metric jitter: 11.967h period, depth-independent
KM3NeT-IC flux ratio	KM3NeT ARCA public + IceCube HESE [3,4]	Phase inversion: variance minimized at dt_geo=5.9836h
UHECR arrival times	Telescope Array + Auger E>57EeV catalogs [5,6]	Phase coherence: energy-independent (rigidity exclusion)
Amaterasu (244EeV)	TA reconstruction + JF12 model [5,7]	Torsional lensing: 3-5 deg residual toward Kadunic Vector direction

4. Five Veto Tests

Veto V1: Ice-Layer Independence

Red Team Attack:	The IceCube angular floor is residual ice scattering. Tilt, birefringence, and seasonal temperature variations produce apparent periodicities.
Hardening:	Ice scattering correlates with vertex depth and detector sector. The Gray Sector metric does not care about ice. Segment events by vertex depth (dust layer vs clear ice) and by detector sector. Fit amplitude and phase independently in each bin.
Accept if:	Amplitude and phase consistent across all depth bins and sectors (χ^2 $p > 0.05$).
Hard Fail if:	**Amplitude varies with depth or phase shifts between sectors: glaciological.**

Veto V2: Geographic Phase-Lock

Red Team Attack:	The 3.5-3.8 sigma KM3-IC tension is a statistical fluctuation. Post hoc corrections can always reduce variance if you try enough shifts.

Hardening:	The shift Δt_{geo} = 5.9836h is fixed a priori. Test against 10,000 random time shifts in [0,24h]. Additionally: a statistical fluke is a one-time event; the Gray Sector predicts a continuous periodic flux oscillation at f_{sig}.
Accept if:	Variance reduction at Δt_{geo} is significant vs random ensemble ($p < 0.001$) AND Lomb-Scargle shows peak at f_{sig} (>3σ).
Hard Fail if:	**Random shifts perform equally well, or no periodic signal in flux ratio.**

Veto V3: Rigidity Exclusion	
Red Team Attack:	UHECR phase coherence is an artifact of galactic magnetic fields, which have complex spatial structure imprinting preferred arrival directions.
Hardening:	Magnetic deflection is rigidity-dependent: $\theta_{mag} \sim Ze/E$. 100 EeV and 200 EeV protons must deflect differently. Torsional coupling is energy-independent. Divide events into 57-100 EeV and >100 EeV bins. Fit sidereal phase ϕ_0 at fixed f_{sig} in each bin.
Accept if:	Phase ϕ_0 consistent across energy bins ($\lvert\phi_0^{high} - \phi_0^{low}\rvert < 10$ deg).
Hard Fail if:	**Phase shifts significantly with energy: magnetic deflection confirmed.**

Veto V4: Solar Null	
Red Team Attack:	Any sidereal modulation is a diurnal artifact: day-night calibration variations, atmospheric muon background, or human activity patterns.
Hardening:	Diurnal effects are locked to 24.00h. The Gray Sector signal is locked to 23.934h and drifts 3.932 min/day. Fold all datasets at $\omega_{solar} = 2\pi/24$h.
Accept if:	Best-fit amplitude in solar fold consistent with zero (<1σ).
Hard Fail if:	**Non-zero solar amplitude: diurnal systematic.**

Veto V5: Torsional Lensing Direction	
Red Team Attack:	Amaterasu trajectory is random scatter. With enough UHECR events, some will appear to come from voids by chance.

Hardening:	Torsional lensing predicts deflection correlated with the Kadunic Vector direction (RA~03h15m, Dec~-19 deg, hypothesis). Events aligned with this direction should show larger source offsets than perpendicular events.
Accept if:	Aligned events show significantly larger source offsets than perpendicular (K-S test $p < 0.01$).
Hard Fail if:	**No angular dependence in source offsets.**

5. Trilogy Completion: Statistical Power

PHASE-LOCKED POWER: All three signal parameters (f_{sig}, ε_0, α_K) are fixed a priori from Part I. No new free parameters are introduced in Parts II or III. The trials factor across all three parts is 1 — not 3 experiments but 3 independent verifications of the same prediction. The probability of a false positive across all three parts simultaneously is $(p_I) \times (p_{II}) \times (p_{III})$, which drops below 10^{-20} for individual p-values of 10^{-7}.

Observable	N events	Key test	Path to 5-sigma
IC angular jitter	~10^4/yr	Depth-independence (V1)	1yr Upgrade data; coherent phase integration at fixed f_sig
KM3-IC phase sync	~100 UHE events	Variance reduction (V2) + periodicity	1yr contemporaneous; Lomb-Scargle + random shift ensemble
UHECR energy-independence	~200 (E>57EeV)	Phase consistency across E bins (V3)	Not an amplitude search; discrimination test at fixed frequency

OMEGA SERIES VOL. I — TRILOGY COMPLETE

PART I: The Kinetic Hiss — Baksan 24yr + SNO+ 685d + IceCube temporal

PART II: Orbital Metric Audit — GNSS DBDs + ISL residuals + Galileo G2

PART III: High-Energy Jitter — IC angular + KM3-IC tension + Amaterasu

One signal: f_sig=2.3212e-05Hz | eps_0=0.12 s/day | alpha_K=1.71e-4

Geographic phase correction: dt_geo=5.9836h = T_sig/2 (pre-registered)

Rigidity discriminant: energy-independent phase excludes magnetic deflection

Trials factor = 1 across all three parts (parameters fixed in Part I)

Random seed: 20260414 | doi:10.5281/zenodo.1082194

The high-energy sky will decide.

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